It is of interest to note that if we assume that the angular resolution of the human eye is primarily due to diffraction effects then it will be given by

$$\Delta\theta \sim \frac{\lambda}{D} \approx \frac{6\times10^{-5}}{2\times10^{-1}} = 3\times10^{-4}\ \text{rad.} \qquad \textbf{(18.40)}$$

where we have assumed the pupil diameter to be 2 mm. Thus, at a distance of 20 m, the eye should be able to resolve two points which are separated by a distance

$$3\times10^{-4}\times20 = 6\times10^{-3}\ \text{m} = 6\ \text{mm}$$

One can indeed verify that this result is qualitatively valid by finding the distance at which the millimetre scale will become blurred.

In the above discussion, we have assumed that the two object points produce identical (but displaced) Airy patterns. If that is not the case then the two central maxima will have different intensities; accordingly one has to set up a modified criterion for the limit of resolution such that the two maxima stand out.

### 18.5.1 Resolving Power of a Microscope

We next consider the resolving power of a microscope objective of diameter $D$ as shown in Fig. 18.28. Let $P$ and $Q$ represent two closely spaced self-luminous point objects which are to be viewed through the microscope. Assuming the absence of any geometrical aberrations, rays emanating from the points $P$ and $Q$ will produce spherical wavefronts (after refraction through the lens) which will form Airy patterns around their paraxial image points $P'$ and $Q'$. For the points $P$ and $Q$ to be just resolved, the point $Q'$ should lie on the first dark ring surrounding the point $P'$ and, therefore, we must have

$$\sin\alpha' \approx \frac{1.22\,\lambda}{D} = \frac{1.22\,\lambda_0}{n'D} \qquad \textbf{(18.41)}$$

where $n$ and $n'$ represent the refractive indices of the object and image spaces, $\lambda_0$ and $\lambda\,(=\lambda_0/n')$ represent the wavelength of light in free space and in the medium of refractive index $n'$ respectively. The angle $\alpha'$ is defined in Fig. 18.28 and we have

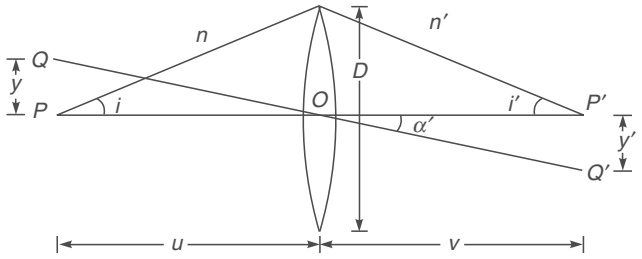



**Fig. 18.28** The resolving power of a microscope objective.

$$\sin\alpha' \approx \frac{y'}{OP'} = \frac{y'\tan i'}{D/2} \approx \frac{y'\sin i'}{D/2} \qquad \textbf{(18.42)}$$

where we have assumed $\sin i' \approx \tan i'$, this is justified since the image distance ($OP'$) is large compared to $D$. Using Eqs. (18.41) and (18.42), we get

$$y' \approx \frac{0.61\,\lambda_0}{n'\sin i'}$$

If we now use the sine law $n'y'\sin i' = ny\sin i$ [see Eq. (4.39) of Chapter 4], we get

$$y \approx \frac{0.61\,\lambda_0}{n\sin i} \qquad \textbf{(18.43)}$$

which represents the smallest distance that the microscope can resolve. The quantity $n\sin i$ is the numerical aperture of the optical system and the resolving power increases with increase in the numerical aperture. It is for this reason that in some microscopes the space between the object and the objective is filled with an oil—and they are referred to as 'oil immersion objectives'. Equation (18.43) also tells us that the resolving power increases with decrease in $\lambda$. As such, one often uses blue light (or even ultraviolet light) for the illumination of the object. For example, in an electron microscope the de Broglie wavelength of electrons accelerated to 100 keV is about $0.03\times10^{-8}$ cm and therefore such a microscope has a very high resolving power.

In the above analysis, we have assumed that the two object points are self-luminous so that the intensities can be added up. However, in actual practice, the objects are illuminated by the same source and, therefore, in general, there is some phase relationship between the waves emanating from the two object points; for such a case the intensities will not be strictly additive (see Sec. 14.6), nevertheless Eq. (18.43) will give the correct order for the limit of resolution.

## 18.6 TWO-SLIT FRAUNHOFER DIFFRACTION PATTERN

LO 6

In Sec. 18.3, we had studied the Fraunhofer diffraction pattern produced by a slit of width $b$ and had found that the intensity distribution consisted of maxima and minima. In this section, we will study the Fraunhofer diffraction pattern produced by two parallel slits (each of width $b$) separated by a distance $d$. We would find that the resultant intensity distribution is a product of the single-slit diffraction pattern and the interference pattern produced by two point sources separated by a distance $d$.

In order to calculate the diffraction pattern we use a method similar to that used for the case of a single slit and assume that the slits consist of a large number of equally spaced point sources and that each point on the slit is a source of Huygens' secondary wavelets. Let the point sources be at $A_1, A_2, A_3, \ldots$ (in the first slit) and at $B_1, B_2, B_3, \ldots$ (in the second slit) [see Fig. 18.29]. As before, we assume that the distance between two consecutive points in either of the slits is $\Delta$. If the diffracted rays make an angle $\theta$ with the normal to the plane of the slits, then the path difference between the disturbances reaching the point $P$ from two consecutive points in a slit will be $\Delta \sin \theta$. The field produced by the first slit at the point $P$ will, therefore, be given by [see Eq. (18.9)]

$$E_1 = A \frac{\sin \beta}{\beta} \cos (\omega t - \beta)$$

Similarly, the second slit will produce a field

$$E_2 = A \frac{\sin \beta}{\beta} \cos (\omega t - \beta - \Phi_1)$$

at the point $P$, where

$$\Phi_1 = \frac{2\pi}{\lambda} d \sin \theta$$

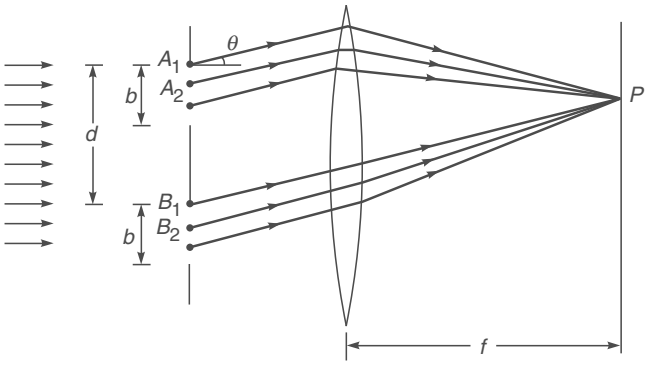



**Fig. 18.29** Fraunhofer diffraction of a plane wave incident normally on a double slit.

represents the phase difference between the disturbances (reaching the point $P$) from two corresponding points on the slits; by corresponding points we imply pairs of points like $(A_1, B_1)$, $(A_2, B_2), \ldots$ which are separated by a distance $d$. Hence the resultant field will be

$$E = E_1 + E_2$$

$$= A \frac{\sin \beta}{\beta} [\cos (\omega t - \beta) + \cos (\omega t - \beta - \Phi_1)]$$

which represents the interference of two waves, each of amplitude $A \frac{\sin \beta}{\beta}$ and differing in phase by $\Phi_1$. The above equation can be rewritten in the form

$$E = A \frac{\sin \beta}{\beta} \cos \gamma \cos \left( \omega t - \frac{1}{2} \beta - \frac{1}{2} \Phi_1 \right)$$

where

$$\gamma = \frac{\Phi_1}{2} = \frac{\pi}{\lambda} d \sin \theta \qquad \textbf{(18.44)}$$

The intensity distribution will be of the form

$$I = 4I_0 \frac{\sin^2 \beta}{\beta^2} \cos^2 \gamma \qquad \textbf{(18.45)}$$

where $I_0 \sin^2 \beta/\beta^2$ represents the intensity distribution produced by one of the slits. As can be seen, the intensity distribution is a product of two terms; the first term ($\sin^2 \beta/\beta^2$) represents the diffraction pattern produced by a single slit of width $b$ and the second term ($\cos^2 \gamma$) represents the interference pattern produced by two point sources separated by a distance $d$. Indeed, if the slit widths are very small (so that there is almost no variation of the $\sin^2 \beta/\beta^2$ term with $\theta$) then one simply obtains the Young's interference pattern (see Sec. 14.6).

In Fig. 18.30, we have shown the two slit diffraction patterns corresponding to $d$ = 0, 0.0176, 0.035 and 0.070 cm with $b$ = 0.0088 cm and $\lambda = 6.328 \times 10^{-5}$ cm. The intensity distribution as predicted by Eq. (18.45) is shown in Fig. 18.31.

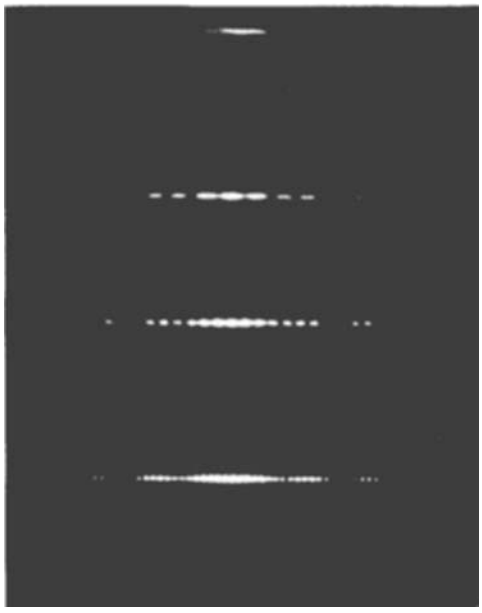


**Fig. 18.30** The double-slit Fraunhofer diffraction pattern corresponding to $b$ = 0.0088 cm and $\lambda = 6.328 \times 10^{-5}$ cm. The values of $d$ are 0, 0.0176, 0.035 and 0.070 cm respectively [Adapted from Ref. 17; used with permission].

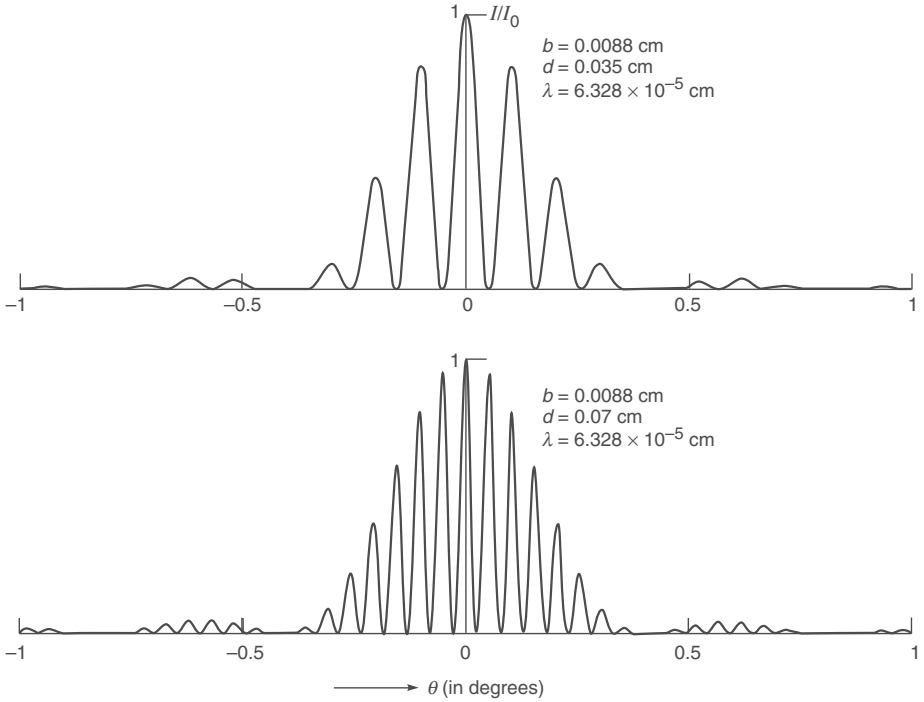



**Fig. 18.31** The double-slit intensity distribution as predicted by Eq. (18.45) corresponding to $d = 0.035$ cm and 0.070 cm, respectively ($b$ = 0.0088 cm and $\lambda = 6.328 \times 10^{-5}$ cm).

### 18.6.1 Positions of Maxima and Minima

Equation (45) tells us that the intensity is zero wherever

$$\beta = \pi, 2\pi, 3\pi, \ldots$$

or when

$$\gamma = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$$

The corresponding angles of diffraction will be given by the following equation:

$$\left.\begin{aligned} b \sin\theta &= m\lambda; \quad (m = 1, 2, 3, \ldots) \\ \text{and} \qquad d \sin\theta &= \left(n + \frac{1}{2}\right)\lambda; \quad (n = 1, 2, 3, \ldots) \end{aligned}\right\} \qquad \textbf{(18.46)}$$

The interference maxima occur when

$$\gamma = 0, \pi, 2\pi, \ldots$$

or when,

$$d \sin\theta = 0, \lambda, 2\lambda, 3\lambda, \ldots \qquad \textbf{(18.47)}$$

The actual positions of the maxima will approximately occur at the above angles provided the variation of the diffraction term is not too rapid. Further, a maximum may not occur at all if $\theta$ corresponds to a diffraction minimum, i.e., if $b \sin\theta = \lambda, 2\lambda, 3\lambda, \ldots$ These are usually referred to as missing orders. For example, in Fig. 18.31 we can see that for $b$ = 0.0088 cm, the interference maxima are extremely weak around $\theta \simeq 0.41^\circ$; this is because of the fact that at

$$\theta = \sin^{-1}\left(\frac{\lambda}{b}\right)$$

$$= \sin^{-1}\left[\frac{6.328 \times 10^{-5}}{8.8 \times 10^{-3}}\right] = \sin^{-1}\left[7.19 \times 10^{-3}\right]$$

$$\simeq 0.00719 \text{ radians}$$

$$\simeq 0.412^\circ$$

the first minimum of the diffraction term occurs.

**Example 18.9** Consider the case when $b = 8.8 \times 10^{-3}$ cm, $d = 7.0 \times 10^{-2}$ cm and $\lambda = 6.328 \times 10^{-5}$ cm (see Fig. 18.31). How many interference minima will occur between the two diffraction minima on either side of the central maximum? In the experimental arrangement corresponding to Fig. 18.30 the screen was placed at a distance of 15′. Calculate the fringe width.

*Solution:* The interference minima will occur when Eq. (18.46) is satisfied, i.e., when

$$\sin\theta = \left(n + \frac{1}{2}\right)\frac{\lambda}{d} = 0.904 \times 10^{-3}\left(n + \frac{1}{2}\right);$$

$$n = 0, 1, 2, \ldots$$

$$= 0.452 \times 10^{-3}, 1.356 \times 10^{-3}, 2.260 \times 10^{-3},$$

$$3.164 \times 10^{-3}, 4.068 \times 10^{-3}, 4.972 \times 10^{-3},$$

$$5.876 \times 10^{-3}, 6.780 \times 10^{-3}$$